

MANAGEMENT REVIEW

Management without theatre.

KPIS

KPIS THAT WARN YOU IN TIME

Why a result tells you what has already happened, which measures show what is coming, and how to pair the two.



60%

of managers say their KPIs need to improve.

MIT SMR & BCG, 2024

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A KPI that turns red has already done its damage. This issue asks a practical question: which measures show a problem while there is still time to act?

Kaplan and Norton built the Balanced Scorecard on the idea that results and the drivers of results have to be measured together. OSHA asks the same of safety: count the injuries, but first measure what prevents them. Surveys of managers show the gap that remains: most say their KPIs need to improve, and only about a quarter of senior managers strongly agree that their KPIs are tied to strategy.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

Practice

The steps and the card are proposals to try, not research results.

COVER STORY

The result always arrives late

A lagging indicator counts what has already happened. A leading indicator measures what makes it happen. You need both, but only one of them gives you time.

Leading

Proactive, preventive, predictive

Shows whether the work that prevents problems is being done

Example: share of workers at the refresher training

Lagging

Counts events that have already happened

Shows whether the effort worked

Example: the rate of injuries and illnesses

This is how OSHA, the US workplace safety agency, describes them in its 2019 guide, and it advises using both. Kaplan and Norton made the same argument for the whole business in 1992: financial measures show the results of actions already taken, so the Balanced Scorecard adds measures of customers, internal processes, and innovation and improvement, which drive the results to come.



1975

Goodhart: a statistic tends to break down once it is used for control.



1992

Kaplan and Norton: the Balanced Scorecard, with four perspectives.



1996

Outcome measures, paired with the drivers that produce them.



2019

OSHA's guide to leading indicators for safety.

OUR READING

A lagging KPI tells you whether you made it. A leading one tells you whether you will make it, while you can still change the answer.

Sources: OSHA, 2019; Robert S. Kaplan & David P. Norton, Harvard Business Review, 1992; Robert S. Kaplan & David P. Norton, Harvard Business Review, 1996; Charles Goodhart, 1975

THE NUMBERS

KPIs that fall short

Two surveys of managers, six years apart, describe the same gap: the KPIs are there, the confidence in them is not.

MANAGERS AND THEIR KPIS, 2024

Say their KPIs need to improve

60%

Use AI to create new KPIs

34%

26%

of senior managers strongly agree that their KPIs are aligned with their organisation's strategic objectives (2018).

3,000+

people surveyed for the 2024 report, in 25+ industries and 100 countries

90%

of those who use AI for KPIs say their KPIs improved

3,200+

people surveyed for the 2018 study

OUR READING

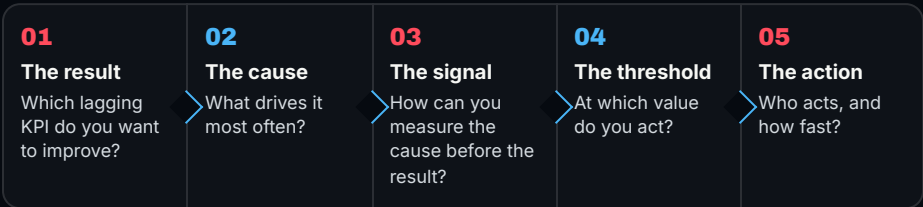
If only a quarter of senior managers are fully convinced that their KPIs follow the strategy, the problem is not a shortage of numbers. It is that the numbers do not say early enough what matters.

Sources: MIT Sloan Management Review & BCG, 2024; MIT Sloan Management Review & Google, 2018

THE MODEL

From the result back to the cause

A leading indicator is not chosen from a list. It is found by walking back from the result you want to the work that produces it.



HYPOTHETICAL EXAMPLE, A WAREHOUSE

The result	Incomplete orders reported by customers.
The cause	Items missing when the truck is loaded.
The signal	The share of orders fully scanned before loading, per shift.
The threshold	Below 98% in a shift.
The action	The shift lead checks the open orders with the team before departure.

The numbers are invented to show the method.

OUR READING

The best leading indicator is close to the work: the team sees it during the shift and can change it the same day.

Source: Robert S. Kaplan & David P. Norton, Harvard Business Review, 1996

The chain is the editors' model, built on the logic of Kaplan and Norton (1996): outcome measures and the drivers that produce them.

HOW TO START

Three ways to a first indicator

OSHA's guide suggests three starting points. Its examples come from safety, but the logic works for any process.

1

From data you already collect

For example, the share of workers who attend the refresher training.

2

From a hazard you know

For example, walkways inspected and cleared, truck brakes replaced on time, how long the lift team takes to arrive.

3

From one part of the programme

For example, preventive maintenance done on schedule, or hazards corrected within the day, week or month they were found.

“When a measure becomes a target, it ceases to be a good measure.”

Marilyn Strathern's 1997 wording of an idea the economist Charles Goodhart put forward in 1975.

OUR READING

A leading indicator is for learning, not for punishing. If it becomes a target at any price, people learn to fill the number instead of fixing the cause.

Sources: OSHA, 2019; Marilyn Strathern, European Review, 1997; Charles Goodhart, 1975

HOW IT IS MEASURED

Five tests for a leading indicator

Before you put a new indicator on the board, check it against five questions. If it fails one, it will not warn you in time.

01 It moves first

When it worsens, does the result usually worsen later?

02 The team can move it

Can the people who see it change it with their own work?

03 It is fast

Is it ready in the same shift or day, not at the end of the month?

04 It has a threshold and an owner

Is it clear at which value someone acts, and who?

05 It is reviewed

After three months, did it really move before the result?

OUR READING

An indicator that does not pass the fifth test is a guess. Keep the ones that proved they warn, drop the ones that only add work.

The five tests are a practice proposed by the editors.

[More in the essay](#)[KPIs the team trusts](#)

TOOL OF THE ISSUE

The card for one leading indicator

One lagging KPI, one leading indicator for it. Fill it in with the people who do the work, not for them.

01 The result the lagging KPI and its current value

02 The main cause what drives it most often

03 The leading indicator what exactly is counted, and how

04 When it is measured every shift, every day, every week

05 The threshold and the action at which value, who acts, how fast

06 The review date did it move before the result?

A practice proposed by the editors. Do not write names, customer data or confidential figures on the card.

Open the tool: [KPI Diagnostic](#)

stivencatalyst.com/tools/kpi-diagnostic/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Using Leading Indicators to Improve Safety and Health Outcomes

OSHA, 2019

https://www.osha.gov/sites/default/files/OSHA_Leading_Indicators.pdf

The Balanced Scorecard: Measures That Drive Performance

Robert S. Kaplan & David P. Norton, Harvard Business Review, 1992

Using the Balanced Scorecard as a Strategic Management System

Robert S. Kaplan & David P. Norton, Harvard Business Review, 1996

The Future of Strategic Measurement: Enhancing KPIs With AI

MIT Sloan Management Review & BCG, 2024

<https://sloanreview.mit.edu/projects/the-future-of-strategic-measurement-enhancing-kpis-with-ai>

Leading With Next-Generation Key Performance Indicators

MIT Sloan Management Review & Google, 2018

<https://sloanreview.mit.edu/projects/leading-with-next-generation-key-performance-indicators>

Problems of Monetary Management: The U.K. Experience

Charles Goodhart, 1975

'Improving ratings': audit in the British University system

Marilyn Strathern, European Review, 1997

<https://www.cambridge.org/core/journals/european-review/article/improving-ratings-audit-in-the-british-university-system/FC2EE640C0C44E3DB87C29FB666E9AAB>

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked "via". The editors' interpretation is marked "Our reading". Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

stivencatalyst.com/magazine/management-review.html